

**SIMATS ENGINEERING**  
**ASSESSMENT TOOL 1**

**COURSE NAME: MLA03**

**COURSE CODE: Reinforcement learning for Environment and Agent**

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**COURSE OUTCOME COVERED**

*CO1: Analyze reinforcement learning frameworks and Markov Decision Process concepts to model decision-making problems. (BL2, BL3)*

*Assessment Tool Weightage: 40%*

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**Dr. Dumpala Prasanth**

**QUESTIONS: 10**

**Total Marks: 50**

**Annexure A**

**Descriptive Experiment Exam**

**Duration: 2hrs**

**Experiment 1: Installation of Reinforcement Learning Development Environment**

Aim: To install and configure Anaconda Navigator, Python, TensorFlow, Keras, Gymnasium (OpenAI Gym), NumPy, Matplotlib, and other required libraries, and verify the environment by executing a simple Reinforcement Learning program.

**Experiment 2: Familiarization with OpenAI Gym/Gymnasium Environments**

Aim: To create, initialize, and interact with standard Reinforcement Learning environments such as FrozenLake, CartPole, and MountainCar using Gymnasium to understand the concepts of states, actions, rewards, and episodes.

**Experiment 3: Implementation of the Reinforcement Learning Framework**

Aim: To implement the basic Reinforcement Learning framework by designing an agent that interacts with an environment through states, actions, rewards, and policies using Python.

**Experiment 4: Simulation of a Markov Decision Process (MDP)**

Aim: To develop a simple Markov Decision Process model and simulate state transitions, reward functions, and policy execution for understanding sequential decision-making problems.

### **Experiment 5: Bellman Equation Demonstration**

Aim: To implement Bellman Expectation and Bellman Optimality Equations for calculating state-value and action-value functions and analyze the convergence of value estimation.

### **Experiment 6: Exploration vs. Exploitation Strategies**

Aim: To implement  $\epsilon$ -Greedy, Greedy, and Random action-selection strategies and compare their performance in balancing exploration and exploitation during Reinforcement Learning.

### **Experiment 7: Multi-Armed Bandit Problem**

Aim: To implement the Multi-Armed Bandit problem using  $\epsilon$ -Greedy and Upper Confidence Bound (UCB) algorithms and evaluate cumulative rewards obtained by different action-selection methods.

### **Experiment 8: Reward Visualization in Reinforcement Learning**

Aim: To simulate an RL agent in a simple environment and visualize cumulative rewards, episode rewards, and learning performance using Matplotlib graphs.

### **Experiment 9: TensorFlow and Keras-Based Neural Network for RL**

Aim: To build a simple feed-forward neural network using TensorFlow and Keras for approximating value functions in Reinforcement Learning environments.

### **Experiment 10: Mini Reinforcement Learning Project Using CartPole**

Aim: To develop a basic Reinforcement Learning agent using TensorFlow and Keras to solve the CartPole environment and evaluate its learning performance through episode rewards and success rate.

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#### ***Code of Conduct:***

I **D Sai Ramana (192325158L)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

**Signature of the student:** *Sai Ramana*

## RUBRICS FOR CALCULATION

| Criteria                  | Weightage | Level 4 – Exemplary                                         | Level 3 – Proficient                    | Level 2 – Developing                         | Level 1 – Beginning                      |
|---------------------------|-----------|-------------------------------------------------------------|-----------------------------------------|----------------------------------------------|------------------------------------------|
| Understanding of Concepts | 25%       | Demonstrates thorough, accurate understanding of concepts   | Good understanding with minor gaps      | Basic understanding with some misconceptions | Poor understanding; major misconceptions |
| Explanation               | 25%       | Detailed, well-explained answers with relevant examples     | Clear explanation with adequate details | Limited explanation; lacks depth             | Poor or unclear explanation              |
| Application               | 20%       | Effectively applies concepts with strong analytical ability | Applies concepts with minor errors      | Limited application with weak analysis       | No application or incorrect analysis     |
| Organization              | 15%       | Well-structured answers with logical flow and coherence     | Organized with minor issues in flow     | Basic structure, lacks clarity               | Poor organization, incoherent answers    |
| Accuracy                  | 10%       | Completely accurate and covers all required points          | Mostly accurate with minor omissions    | Partially correct with missing points        | Incorrect answers with major gaps        |
| Clarity                   | 5%        | Clear, neat, professional presentation                      | Understandable with minor issues        | Limited clarity, readability issues          | Poorly written, difficult to understand  |

## EXPERIMENT 1: Installation of Reinforcement Learning Development Environment

### Question (Aim):

To install and configure Anaconda Navigator, Python, TensorFlow, Keras, Gymnasium (OpenAI Gym), NumPy, Matplotlib, and other required libraries, and verify the environment by executing a simple Reinforcement Learning program.

### Algorithm:

1. Install Anaconda and Python.
2. Install TensorFlow, Keras, Gymnasium, NumPy, and Matplotlib.
3. Import the required libraries.
4. Verify successful installation.

### Code:

```
import tensorflow as tf
import gymnasium as gym
import numpy as np
import matplotlib.pyplot as plt
print("TensorFlow:", tf.__version__)
print("Gymnasium Installed")
print("NumPy:", np.__version__)
```

### Input & output:

```
Input:
No user input.

Output:
TensorFlow: 2.x.x
Gymnasium Installed
NumPy: 1.x.x
```

### Result:

All the required Reinforcement Learning libraries were installed and verified successfully.

## EXPERIMENT 2: Familiarization with OpenAI Gym/Gymnasium Environments

### Question (Aim):

To create, initialize, and interact with standard Reinforcement Learning environments such as FrozenLake, CartPole, and MountainCar using Gymnasium to understand the concepts of states, actions, rewards, and episodes.

### Algorithm:

1. Create the FrozenLake environment.
2. Reset the environment.
3. Select random actions.
4. Observe states and rewards.
5. Stop when the episode ends.

### Code:

```
import gymnasium as gym
env = gym.make("FrozenLake-v1")
state, _ = env.reset()
for i in range(5):
    action = env.action_space.sample()
    state, reward, done, truncated, info = env.step(action)
    print("State:", state, "Reward:", reward)
    if done or truncated:
        break
env.close()
```

### Input & output:

#### EXPERIMENT 2

##### Input:

No user input.

##### Output:

State: 0 Reward: 0

State: 4 Reward: 0

State: 5 Reward: 1

(or similar values depending on random actions)

### Result:

Successfully interacted with the Gymnasium environment and observed states, actions, and rewards.

## EXPERIMENT 3: Implementation of the Reinforcement Learning Framework

### Question (Aim):

To implement the basic Reinforcement Learning framework by designing an agent that interacts with an environment through states, actions, rewards, and policies using Python.

### Algorithm:

1. Create the environment.
2. Initialize the agent.
3. Select an action.
4. Receive reward.
5. Continue until the episode ends.

### Code:

```
import gymnasium as gym
env = gym.make("CartPole-v1")
state, _ = env.reset()
done = False
while not done:
    action = env.action_space.sample()
    state, reward, done, truncated, info = env.step(action)
    print("Reward:", reward)
    done = done or truncated
env.close()
```

### Input & output:

```
Input:
No user input.

Output:
State: 0 Reward: 0
State: 4 Reward: 0
State: 5 Reward: 1
(or similar values depending on random actions)
```

### Result:

The Reinforcement Learning framework was successfully implemented and the agent interacted with the environment.

## EXPERIMENT 4: Simulation of a Markov Decision Process (MDP)

### Question (Aim):

To develop a simple Markov Decision Process model and simulate state transitions, reward functions, and policy execution for understanding sequential decision-making problems.

### Algorithm:

1. Define states.
2. Assign rewards.
3. Simulate state transitions.
4. Display rewards.

### Code:

```
states = ["A", "B", "C"]
rewards = [5, 10, 15]
for s, r in zip(states, rewards):
    print("State:", s, "Reward:", r)
```

### Input & output:

```
Input:
No user input.

Output:
State: A Reward: 5
State: B Reward: 10
State: C Reward: 15
```

### Result:

The Markov Decision Process was successfully simulated with state transitions and rewards.

## EXPERIMENT 5: Bellman Equation Demonstration

### Question (Aim):

To implement Bellman Expectation and Bellman Optimality Equations for calculating state-value and action-value functions and analyze the convergence of value estimation.

### Algorithm:

1. Initialize the value function.
2. Apply the Bellman equation.

3. Update the value repeatedly.
4. Display the final value.

**Code:**

```
gamma = 0.9
reward = 10
value = 0
for i in range(10):
    value = reward + gamma * value
print("State Value:", value)
```

**Input & output:**

```
Input:
Reward = 10
Gamma = 0.9

Output:
State Value: 65.13215599
(Value may vary depending on the number of iterations.)
```

**Result:**

The Bellman equation successfully estimated the state value.

**EXPERIMENT 6: Exploration vs. Exploitation Strategies****Question (Aim):**

To implement  $\epsilon$ -Greedy, Greedy, and Random action-selection strategies and compare their performance in balancing exploration and exploitation during Reinforcement Learning.

**Algorithm:**

1. Set epsilon.
2. Generate a random number.
3. Explore if random number is less than epsilon.
4. Otherwise exploit.

**Code:**

```
import random
epsilon = 0.2
```



```
for i in range(10):
    if random.random() < epsilon:
        print("Explore")
    else:
        print("Exploit")
```

### **Input & output:**

```
Input:
Epsilon = 0.2

Output:
Exploit
Explore
Exploit
Exploit
Explore
...
(Output varies because of random action selection.)
```

### **Result:**

The exploration and exploitation strategies were demonstrated successfully.

## **EXPERIMENT 7: Multi-Armed Bandit Problem**

### **Question (Aim):**

To implement the Multi-Armed Bandit problem using  $\epsilon$ -Greedy and Upper Confidence Bound (UCB) algorithms and evaluate cumulative rewards obtained by different action-selection methods.

### **Algorithm:**

1. Create bandits.
2. Select an action.
3. Generate reward.
4. Repeat the process.

### **Code:**

```
import numpy as np
bandits = [0.2, 0.5, 0.8]
for i in range(10):
```

```
action = np.random.randint(3)
reward = np.random.binomial(1, bandits[action])
print("Action:", action, "Reward:", reward)
```

### Input & output:

```
Input:
Bandit Probabilities = [0.2, 0.5, 0.8]

Output:
Action: 2 Reward: 1
Action: 0 Reward: 0
Action: 1 Reward: 1
...
(Output varies for each execution.)
```

### Result:

The Multi-Armed Bandit problem was successfully simulated and rewards were generated.

## EXPERIMENT 8: Reward Visualization in Reinforcement Learning

### Question (Aim):

To simulate an RL agent in a simple environment and visualize cumulative rewards, episode rewards, and learning performance using Matplotlib graphs.

### Algorithm:

1. Store rewards.
2. Plot the graph.
3. Display the graph.

### Code:

```
import matplotlib.pyplot as plt
rewards = [1,2,3,4,5,6,7,8,9,10]
plt.plot(rewards)
plt.xlabel("Episode")
plt.ylabel("Reward")
plt.title("Reward vs Episode")
plt.show()
```

### Input & output:

```
Input:
Rewards = [1,2,3,4,5,6,7,8,9,10]

Output:
A line graph showing Reward vs Episode is displayed.
```

**Result:**

The reward graph was successfully displayed using Matplotlib.

**EXPERIMENT 9: TensorFlow and Keras-Based Neural Network for Reinforcement Learning****Question (Aim):**

To build a simple feed-forward neural network using TensorFlow and Keras for approximating value functions in Reinforcement Learning environments.

**Algorithm:**

1. Create a Sequential model.
2. Add Dense layers.
3. Compile the model.
4. Display the model summary.

**Code:**

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

model = Sequential([
    Dense(24, activation='relu', input_shape=(4,)),
    Dense(24, activation='relu'),
    Dense(2)
])

model.compile(optimizer='adam', loss='mse')
model.summary()
```

**Input & output:**

```
Input:
Input Shape = (4,)
Hidden Layers = 24, 24
Output Layer = 2

Output:
Model: "sequential"
Dense (24)
Dense (24)
Dense (2)
Total Parameters: ...
(Model summary is displayed.)
```

**Result:**

The feed-forward neural network was successfully created and compiled.

**EXPERIMENT 10: Mini Reinforcement Learning Project Using CartPole****Question (Aim):**

To develop a basic Reinforcement Learning agent using TensorFlow and Keras to solve the CartPole environment and evaluate its learning performance through episode rewards and success rate.

**Algorithm:**

1. Create the CartPole environment.
2. Reset the environment.
3. Select random actions.
4. Calculate the total reward.
5. Display the final reward.

**Code:**

```
import gymnasium as gym
env = gym.make("CartPole-v1")
state, _ = env.reset()
total_reward = 0
done = False
while not done:
    action = env.action_space.sample()
    state, reward, done, truncated, info = env.step(action)
```

```
total_reward += reward  
done = done or truncated  
print("Total Reward:", total_reward)  
env.close()
```

### **Input & output:**

```
Input:  
No user input.  
  
Output:  
Total Reward: 25.0  
(or any value depending on the episode performance.)
```

### **Result:**

The CartPole Reinforcement Learning agent successfully interacted with the environment, and the total episode reward was obtained.